

dynamic games without the agonizing pain*

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This probably won't be comprehensive. I'm trying to cover the big ideas Elliot brought up in class that I had a hard time understanding, which is most of them.

Feel free to add notes, edit, or share!

1 Definitions

For definitions, see section 6.1 of [Osborne and Rubinstein](#).

1.1 Intuition

- Think of a **strategy** as “what would I do at every node I could be called on to act?”
 - Or a function from non-terminal histories to actions.
 - At a terminal history I can't be called to act. The same goes for nodes where it's not my turn. (More formally my action set is a singleton, e.g. {stand there}.)
- We would like to be able to say “at this node in the game tree”. Instead of talking about nodes, let's talk about **histories**.
 - A history is a sequence of actions where it is possible to take all of them in order by walking down the game tree.
 - When I say history h , I'm really saying “this node in the game tree you arrive at, if you took the sequence of actions defined by history h ”.

1.2 Questions

- I have no idea what the action set $\mathcal{A}$ does, or what it looks like.
 - Maybe it makes it easier to define what a history is. A history is just a sequence of actions, so the space of length- t histories is something like $H_t \subseteq (\mathcal{A}^N)^t$.
 - Take this example (the “holdup game”).

*title from [An Introduction to the Conjugate Gradient Method Without the Agonizing Pain](#)

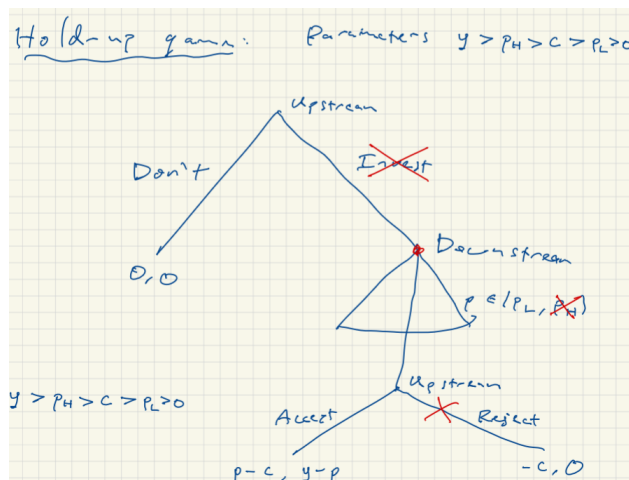


Figure 1: Q: Is $\mathcal{A} = \{invest, don't, accept, reject\} \times \{p_L, p_H\}$? Or is it the union, so $\mathcal{A} = \{invest, don't, accept, reject\} \cup \{p_L, p_H\}$?

A: Upon further thought I think it's the union of all possible actions.

2 IDSDS and backward induction

2.1 Intuition

- **Strict domination:** A strategy is strictly dominated (in a dynamic game) if it is strictly dominated at *any history (!)* by another strategy.
 - *These histories need not be reached!* A strategy can be dominated “off the equilibrium path of play”. It’s still a bad idea to play such a strategy because you can always find a better one.
- **Backwards induction**
 - As a definition: just apply IDSDS, using our notion of domination as “domination at any history”.
 - As a procedure: start at the last node, figure out the best response. Given those best responses (“continuation values”), figure out what the best responses at the second to last node are, and so on.
 - Comment: *BI is really strong.* Actually all BI solutions are (subgame perfect) Nash equilibria.
 - * When you do BI, first you’re figuring out best responses at the last node, then best responses at the second to last node, and so on.
 - * So we’ve found best responses in every subgame, because we’ve worked up the tree, going through every node (subgame).
 - BI is very strong because our notion of strict domination rules out lots of strategies.
 - * If a strategy is dominated at the initial history, it must be worse than another strategy, *no matter what the other players do*. Very few strategies are this bad.
 - * If a strategy is strictly dominated at the last (non-terminal) node, it’s just *not a best response*. If we’re at the last node, the other player has no more choices to make! Many strategies are not best responses, so we can delete lots of stuff.

3 Extensive Form Rationalizability

3.1 Intuition

- A strategy **allows** a history h , if we can start at the initial history, play the strategy, and end up at history h .

- **Building blocks:** As before, we want to think of rationalizability in terms of strategies that are best responses to beliefs. Let's define what beliefs and best responses are in dynamic games.
- **Conditional belief system:** It's my belief about what the other players are doing at every non-terminal node. Beliefs are probability distributions over the strategies other players could be taking.

– Not every belief can be a CBS. A CBS is a logically consistent belief. (The following two conditions say what logical consistency requires.)

* We define beliefs in this complicated way because we only want to think about responding to sensible beliefs.

1. If I reach a history, I must believe with probability one that people played strategies that lead to this history. Also, I must believe with probability zero people played strategies inconsistent with this history.

– Suppose I'm playing ping pong and the ball is flying towards me. I must believe with probability one the other person hit the ball back at me and with probability zero that they missed.

2. (Sometimes called Bayes consistency.) Suppose I was at history h , and now I reach history h' . What should I believe at h' ?

– Put probability zero on any strategy inconsistent with h' . That's the above restriction.

– For strategies that lead to h' , I update my beliefs using Bayes' rule.

– Elliot gives

$$p_i^h(s_{-i}) = p_i^h(S_{-i}(h')) p_i^{h'}(s_{-i}).$$

It's easier if we rearrange to get

$$p_i^{h'}(s_{-i}) = \frac{p_i^h(s_{-i})}{p_i^h(S_{-i}(h'))}.$$

The left hand side says "what's the probability the other players are playing s_{-i} ?"

The right hand side is Bayes' rule, namely

$$P[\text{others play } s_{-i} | \text{reach } h'] = \frac{P[\text{others play } s_{-i} \text{ AND we reach } h']}{P[\text{we reach } h']} = \frac{p_i^h(s_{-i})}{p_i^h(S_{-i}(h'))}.$$

My new beliefs at h' are proportional to my old beliefs at h .

* The numerator of the RHS are my old beliefs at h (at least, for the strategies that allow h'), and the denominator just rescales them.

- **Sequential best reply:** A (pure) strategy is a sequential best reply to a belief system if it best responds to that belief system *at every history*.

– Again, I literally mean at every history, including ones that won't be reached!

* Each h is a node where I could be potentially asked to act. I always want to act optimally.

– The definition requires the belief to be a conditional belief system. We want to have a different belief at every history, and these beliefs need to be logically consistent.

* Returning to ping pong, if the ball is flying at me, the opponent must have hit the ball. If the ball is flying toward the opponent, I must have hit it.

– **Mathematically,** pure strategy $s_i \in S_i$ is a sequential best reply to a conditional belief system p_i if at every non-terminal history $h \in H \setminus Z$

$$s_i \in \arg \max_{\tilde{s}_i \in S_i} \sum_{s_{-i} \in S_{-i}} p_i^h(s_{-i}) u_i(\tilde{s}_i, s_{-i} | h)$$

- * In English, if I believe p_i about what the other players will do, a sequential best reply maximizes my expected utility against p_i at every history.
 - Read $\sum_{s_{-i} \in S_{-i}} p_i^h(s_{-i}) u_i(\tilde{s}_i, s_{-i}|h) \equiv \mathbb{E}_{s_{-i} \sim p_i}[u_i(\tilde{s}_i, s_{-i}|h)]$ as “my expected utility when my opponents are playing the mixed strategy p_i ”.
- If I believed my opponent will for sure take a single action at each h , then this definition reduces to “best responding at every h ”.
- **Rationalizable set:** Use something like iterated deletion to find the rationalizable set.

For each player i : start with their original set of strategies. Set $S_i^{(0)} = S_i$.

For each step k :

- Only keep strategies that best respond to some belief (CBS). Call the remaining strategies $S_i^{(k)}$.
- Update the player’s beliefs for each non-terminal history. The player must believe that the surviving strategies from the last round are played with probability one.
 - * (If the surviving strategies do not allow history h , then the player can believe anything about what happens next.^a)

Until you converge on a set of strategies $S_i^{(\infty)}$, which are the rationalizable strategies for player i .

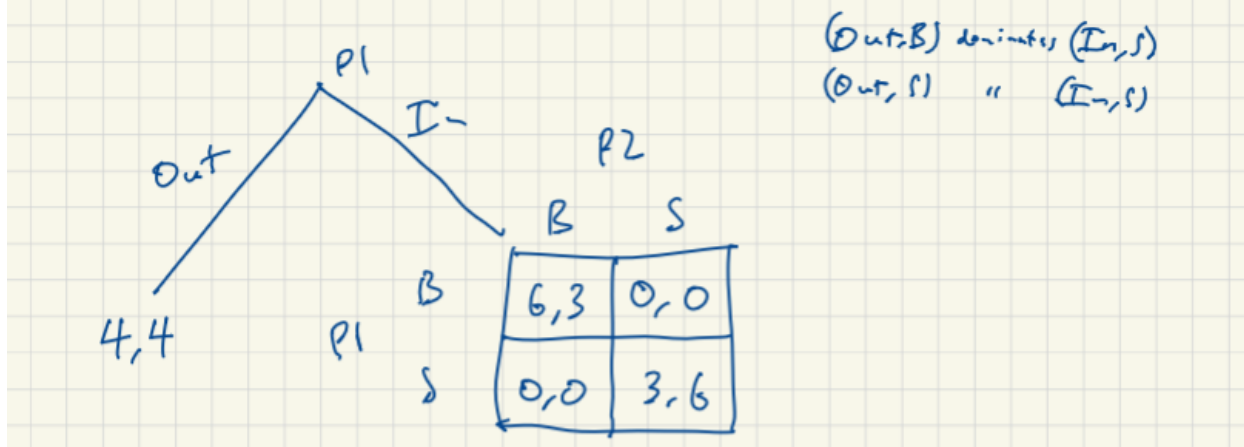
^aTo be more precise, the player has a CBS, so they must believe that the strategies played lead to h , but those strategies could play in any arbitrary way after h .

- Do not be alarmed that this is a complicated, iterative process. We’re answering “what would you play if you knew that I knew that you knew... that I rationally best reply”? It’s a complicated question with many layers.
- In practice, at every step, you’ll need to check *both* that you’re throwing out never-best-replies *and* you’re keeping best replies.

3.2 Questions

- EFR strategy profiles are not necessarily (subgame perfect) Nash equilibria, and vice versa. I don’t have good intuition for why. (Remember backward induction yields (subgame perfect) Nash equilibria).
- The game below. We said in class it has no backward induction solution. But it has a forward induction solution, because (In, S) is strictly dominated, so if P2 sees In , they infer that P1 is playing (In, B) and P2 plays B .
 - This is meant to illustrate the difference between backward induction and forward induction. Forward induction is meant to be an example of extensive form rationalizability.
 - * We defined “backward induction solution” to mean the limit of IDSDS, not whatever one gets by following the procedure of working back up the tree.
 - * Isn’t forward induction also IDSDS?
 - Forward induction means that you’re using your actions to signal your intentions. By going In , P1 signals they intend B . This signaling is what “forward” means (you’re signaling forward to the next move).

Extensive-form rationalizability



- Elliot's explanation: Forward induction / EFR keeps best responses to some belief. Backward induction / IDSDS deletes strictly dominated strategies.
 - Two crucial differences:
 - (a) Never-best-responses are not necessarily strictly dominated because we're looking only at pure strategies.
 - The Lemma from static IDSDS/rationalizability doesn't carry over.
 - (b) **Beliefs are doing a lot of work.** Beliefs restrict what I think the other player can play. (Explained below.)
 - **Forward induction / EFR** in this example:
 - * (In, S) is never a best reply.
 - * Then P2 believes P1 plays $\{ (Out, B), (Out, S), (In, B) \}$.
 - * If we reach BoS, **P2 believes that In was played** (that's what a CBS requires!)
 - **P2 can't make this inference with IDSDS, because (Out, S) survives.**
 - * So P2 plays B.
 - **Backward induction / IDSDS / SPNE** in this example:
 - * (In, S) is strictly dominated by (Out, S) and (Out, B) at the initial history.
 - (No matter what P2 does, playing out guarantees the 4.)
 - * **(Out, S) is not strictly dominated.**
 - It best responds to S at both the initial histories and after "In".
 - * **Then S is never deleted for P2, because it best responds to (Out, S) !**
 - There's no constraint on P2's beliefs to believe "In" after they reach BoS. **That's what adding a CBS buys us.**

4 Subgame Perfect Nash Equilibrium

4.1 Intuition

- A **subgame** is the game starting from a non-terminal history.
- A **subgame perfect Nash equilibrium** is a strategy profile that is a Nash in every subgame.

- That is, each player is best respond to every other player, in every history, including ones that aren't reached.
- The whole game is a subgame of itself, so in static games, SPNE is just NE.
- In the dynamic setting, subgame perfection rules out non-credible threats.

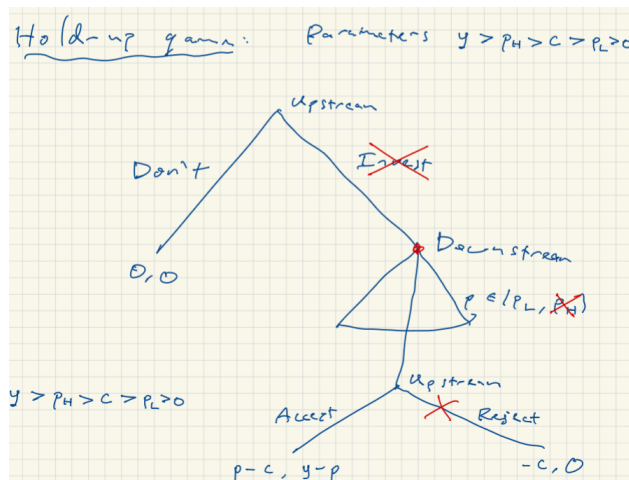


Figure 2: ((Invest, accept p_H , reject p_L), p_H) is a NE, but it relies on the non-credible threat that Upstream will reject p_L . Subgame perfection rules out these non-credible threats.

- Take a strategy s_i . A profitable **one-shot deviation** (OSD) s'_i differs from s_i at only one history h , and is strictly better for player i .
 - It's still a profitable OSD if the deviation is at an off-path h that will never be reached!
 - * The whole point of SPNE is to rule out non-credible threats.
 - Think of s_i and s'_i as *functions* from histories to actions. At each h , they give a rule for how to choose the action to take.
 - To differ only at h means that h is the only place where the *rule* differs.
 - The *action* can differ in many places.
 - * Take the strategy s_i = “play 0 if you’ve always played 0, play 1 if you’ve ever played 1”.
 - * A one-shot deviation is s'_i = “ s_i but play 1 at the initial history”.
 - * Then s_i will lead to many zeros being played and s'_i many ones, even though their rules only differ in one place.
- A game has the **OSD principle** if every strategy profile with no profitable OSD is a SPNE.
- Every finite game has the OSD principle.
 - See the lecture notes for the proof.
 - The idea is that for any deviation at finitely many h , look at the deviation at the last h . Either that deviation is a profitable OSD, or we can just deviate at all the nodes strictly before h . Then we can inductively shrink the deviation until we reach a OSD.
- One can compute SPNE via backward induction.
 - Actually backward induction solutions (“unique limit of IDSDS”) is a subset of SPNE solutions.¹
 - Two examples of SPNE which are not BI solutions:

¹We say that BI is a “refinement” of SPNE.

- * When there are multiple SPNE, there will not be a unique limit of IDSDS, so there will not be a BI solution.
- * The ultimatum game. The unique SPNE is that the first player keeps the entire pie and the second player accepts. But for the first player to have a best response, the second player must accept, so the first player's incentives constrain the second player's actions! BI doesn't handle this sort of reasoning.
- EFR and SPNE are separate solution concepts, and a strategy profile can be an SPNE but not rationalizable and vice versa.

5 Repeated Games

5.1 Intuition

- Repeated games with discounting have the OSD principle.
 - This is because the payoffs are “continuous at infinity”.
 - The payoffs far into the future are discounted so much we can effectively truncate the game to turn it into a finite-horizon one.
 - Counterexamples to the OSD principle require payoffs that count whether someone does something infinitely many times, to maintain a discontinuity at infinity.